MathWorks Math Modeling Challenge 2026

The Rise of Online Gambling: *What's at Stake*

Team #19336

1 Executive Summary

To regulatory authorities and policymakers in the United Kingdom and United States,

In recent years, online gambling, or more specifically sports betting, has seen a major increase following the Murphy v. NCAA decision[6] and the increased accessibility due to the surge of online betting applications[5]. In the UK, sports betting has been more regulated, but has still seen a consistent growth over the years. As a result, both countries have seen substantial growth in government revenue and economic activity[22, 25], but at the same time have stirred uncertainty in regards to addiction and financial distress caused by excessive gambling[]. As policymakers evaluate the effects of sports betting on society, it's important to consider both the fiscal and economic effects that changes can have.

In the first section, we defined the disposable income that the average person had available to them based on relevant demographics, such as age and region. Disposable income varies significantly by age and country, declining through middle age in the U.S. due to higher taxes and essential costs, while remaining more stable and generally higher in the UK due to structural tax differences. As gross income increases, disposable income rises nonlinearly in both countries, though the UK consistently retains a higher share at comparable income levels, highlighting the impact of national tax and expenditure systems. Sensitivity analysis further shows that education level, state of residence, and number of earners influence disposable income more than age alone, demonstrating that financial capacity depends on demographic and structural factors beyond salary.

Subsequently, people bet more based on how much disposable income they have available to them. In the second section, we utilized our definition of disposable income with supporting data[] to figure out the amount of money that people were wasting on sports betting and related activities. We assume the survey data is representative and apply demographic probabilities uniformly, treating key behavioral traits as independent for simplicity. Monthly wagers are estimated using tiered spending ranges, with high spenders modeled using a lognormal distribution, and betting intensity scales with behavioral risk factors and risk-adjusted hold rates. Individuals who do not actively bet have zero impact, and annual outcomes are modeled as evenly distributed across 12 identical months with random variation around expected losses.

However, this information isn't useful until we put it into context. We started with our data from the first two questions in order to gauge total losses caused by gambling, Then, we used a model which estimated the net impact of sports betting on GDP by combining industry production, government spending from tax revenue, and the portion of consumer spending that represents new economic activity rather than reallocated spending. This model was used to show potential benefits of gambling, specifically on the economy in order to justify any potential negatives depicted before. With these models, we saw that the GDP benefits of sports betting were negligible compared to the net loss.

We hope that our findings can educate these esteemed authorities and guide them to making a definitive decision about sports gambling's place within society.

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2 Playing With House Money

2.1 Defining the Problem:

We develop a model to estimate an individual's disposable income based on gross annual income, age, location, education level, and number of earners. Disposable income is defined as income remaining after taxes and essential expenditures, allowing comparison across demographic groups and between the United States and the United Kingdom.

2.2 Assumptions:

1. *The data set provided by M3 is a representative data set of the US and UK as a whole.* For simplicity, we are generalizing the data provided [] for the entirety of each country, however we do reference other data for parts that aren't generalizable, such as specific parameters. [6]

2. *Disposable income (DI) is defined as the portion of total income (TI) remaining after paying essential expenditures.* Defining our DI this way aligns with most economic models, and allows for simplification of our models.

3. *Essential expenditures consist of housing, food/drink, clothing, housing goods, health, transport, communication, and education.* We chose these parameters to be our necessary financial obligations that each individual contributes to.

4. *Total income is defined as the average income remaining after taxes within each region.* Due to the high specificity of local tax laws within regions, we will be getting averages across states, and then averaging those to get region-based incomes. [1, 5, 7]

5. *We define the US regions as such- Northeast: CT, ME, MA, NH, RI, VT, NJ, NY, PA; Midwest: IL, IN, MI, OH, WI, IA, KS, MN, MO, NE, ND, SD; South: AL, AR, DE, FL, GA, KY, LA, MD, MS, NC, OK, SC, TN, TX, VA, WV; West: AK, AZ, CA, CO, HI, ID, MT, NV, NM, OR, UT, WA, WY* [5,7]

6. *In terms of taxes, assume they are filed jointly, there are no child credits, no itemized deductions, and all other deductions come from 2024.* Since the majority of our data is from 2024, for consistency we use tax and deduction information[5] from the same year.

7. Essential expenditures are modeled using demographic group averages rather than individual-level microdata.

8. Demographic adjustments to essential expenditures are treated as additive and independent, with no interaction effects between variables.

9. Currency comparisons between the United States and United Kingdom are not adjusted for purchasing power parity.

10. The model excludes debt obligations, savings behavior, and other non-essential financial flows.

11. Annual income is assumed to be stable with no intra-year volatility.

2.3 Variables

Table 1: Variable symbols, definitions, and units used in the model

Symbol	Definition	Unit	Value
IG	Gross annual income	\$ (US) / £ (UK)	Input
A	Age	Years	Input
L	Location	Categorical	Input
Edu	Education level	Categorical	Input
Ne	Number of earners in household	Count	Input
Tf	Federal income tax (US only)	Dollars (\$)	Computed
Ts	State income tax (US only)	Dollars (\$)	Computed
Tp	Payroll tax (US only)	Dollars (\$)	Computed
Tuk	Income tax (UK only)	Pounds (£)	Computed
NI	National Insurance [30] (UK only)	Pounds (£)	Computed
$Inet$	After-tax income	\$ / £	Computed
$Eess$	Essential expenditures	\$ / £	Computed
DI	Disposable income	\$ / £	Output

2.4 The Model

2.4.1 Model Development

To quantify the financial impact of online gambling, we develop a model to estimate an individual's disposable income. Disposable income is defined as the portion of income remaining after taxes and essential expenditures. We express this relationship as:

$$DI = I_{net} - E_{ess}.$$

Gross annual income IG serves as the primary input variable. Because tax obligations differ between the United States and the United Kingdom, we construct separate tax modules for each country.

For the United States, after-tax income is calculated as:

$$I_{net} = IG - (T_f + T_s + T_p),$$

where T_f represents federal income tax, T_s represents state income tax, and T_p represents payroll tax (Social Security and Medicare). Federal taxes follow the 2024 progressive marginal tax brackets [1]. State taxes are modeled according to either flat or progressive structures depending on state. Payroll taxes include Social Security contributions subject to a wage cap and Medicare contributions with an additional high-income surtax. [9]

For the United Kingdom, after-tax income is computed as:

$$I_{net} = IG - (T_{uk} + NI),$$

where T_{uk} [2] denotes income tax under the 2024/25 marginal rate system and NI denotes National Insurance contributions. Income tax [3] incorporates a personal allowance and progressive marginal rates.

Essential expenditures are modeled using demographic-adjusted expenditure means derived from national consumer expenditure data. Essential spending is defined as the sum of expenditures on food, housing, and healthcare, etc. We model essential expenditures as:

$$E_{ess} = \mu_{income} + \Delta_{age} + \Delta_{location} + \Delta_{education} + \Delta_{earners},$$

where μ_{income} represents baseline essential spending associated with income level, and the Δ terms represent demographic adjustments based on age, geographic location, education level, and number of earners in the household.

Finally, disposable income is constrained to be non-negative:

$$DI = \max(0, I_{net} - E_{ess}).$$

This framework allows us to compare disposable income across demographic groups and across countries while accounting for tax systems and variation in essential spending patterns.

Figure 2.4.1 (Parameter Sensitivity / Tornado Chart): This figure quantifies how sensitive the model output is to each demographic parameter by varying one input at a time from a baseline profile (holding all other inputs fixed). Each horizontal bar shows the change in disposable income ΔDI when a single factor is adjusted to a plausible alternative (for example, changing state, changing education level, changing number of earners, or shifting age group). Larger absolute bars indicate that the corresponding variable has a stronger effect on the final disposable income estimate. In our model, structural and household factors (education level, state/region, and number of earners) produce larger changes in DI than age alone, indicating that taxes and household structure are major drivers of financial flexibility.

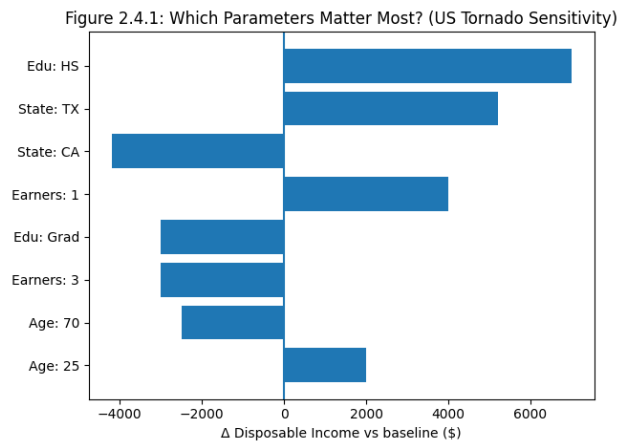

Figure 2.4.1: Which Parameters Matter Most? (US Tornado Sensitivity)

2.4.2 Model Testing and Interpretation

To evaluate both correctness and stability of the disposable income model, we conducted structured testing across controlled demographic scenarios in both the United States and the United Kingdom. All tests were designed to isolate one variable at a time while holding other parameters constant, ensuring interpretability of observed trends.

For income-scaling validation, we fixed age at 40, education at "Some_college," and number of earners at 2. We then evaluated gross incomes ranging from $25,000 to $200,000 in $5,000 increments for a representative U.S. state (Illinois) and U.K. nation (England). This test verifies that disposable income increases monotonically with gross income and reflects progressive taxation structures.

For age-based analysis, we fixed gross income at $85,000 (or equivalent) and computed disposable income share (DI/IG) across the defined age groups. Education and number of earners were held constant to isolate the effect of age-based expenditure adjustments. In the U.S., we tested four representative states—New York (Northeast), Illinois (Midwest), Texas (South), and California (West)—to capture regional variation. In the U.K., we tested England, Wales, Scotland, and Northern Ireland.

For parameter sensitivity testing, we defined a baseline U.S. profile (income $85,000, age 40, Illinois, Some college, 2 earners) and varied one parameter at a time to plausible alternatives while holding all other inputs fixed. The resulting change in disposable income (ΔDI) was recorded for each variation.

Figure 2.4.2 (Disposable Income vs. Gross Income: U.S. vs U.K.):
This figure confirms that disposable income increases with gross income in both systems. The relationship is nonlinear due to progressive marginal tax brackets and structured essential expenditure adjustments. The slope difference between the U.S. and U.K. curves reflects structural differences in tax policy and modeled

spending baselines. This test validates the internal consistency of the tax modules and confirms that the model behaves economically as expected.

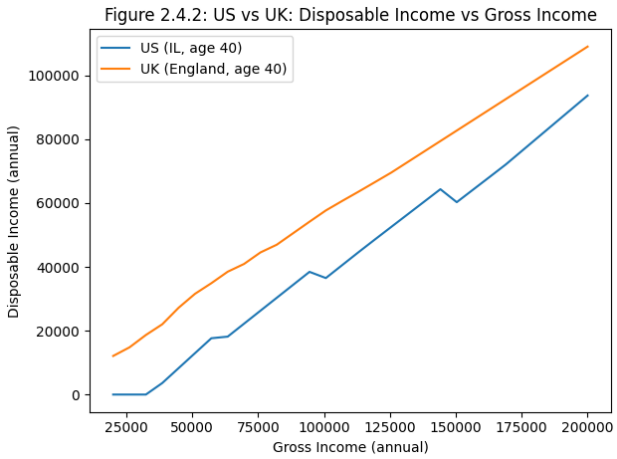

Figure 2.4.2: US vs UK: Disposable Income vs Gross Income

These structured tests demonstrate that the model is monotonic in income, responsive to demographic variation, and capable of distinguishing cross-country structural differences. The trends observed are consistent with economic intuition, supporting the validity of the framework.

2.5 Results

We evaluated the model using a fixed gross annual income of $85,000, holding education level and number of earners constant, while varying age and geographic location. Age groups from under 25 through 75+ were analyzed, with New York, Illinois, Texas, and California representing the four U.S. regions and England, Wales, Scotland, and Northern Ireland representing the United Kingdom. The results show that disposable income share declines through middle age, reflecting higher essential expenditures and cumulative tax burden, and then stabilizes in later years. Variation across U.S. states is more pronounced due to differences in state income taxation, while UK nations exhibit more uniform outcomes. Additionally, disposable income increases nonlinearly with gross income, consistent with progressive tax structures in both countries.

Figure 2.5.1: Heat wave temperatures over 24 hours for the sample data

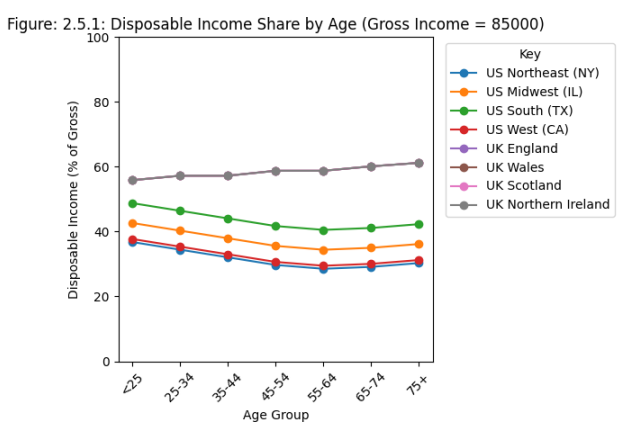

Figure 2.5.1: Disposable Income Share by Age (Gross Income = 85000)

2.6 Discussion

The results highlight that disposable income is driven not only by gross earnings, but by the interaction between taxation structures and essential expenditure patterns. While disposable income increases as gross income rises, the rate of increase slows due to progressive taxation in both countries. Age patterns reveal that disposable income share declines during middle age, reflecting higher housing, healthcare, and household expenditures, before stabilizing in later years. Regional variation is more pronounced in the United States because of state-level income tax differences, with no-tax states exhibiting higher disposable income shares. In contrast, outcomes across the United Kingdom are more uniform due to centralized tax policy. These findings demonstrate that financial flexibility is shaped by institutional structure and household composition as much as by income level itself.

2.7 Sensitivity Analysis

To evaluate the robustness of our disposable income model, we performed a one-at-a-time sensitivity analysis using a fixed baseline profile. The baseline consisted of a gross annual income of $85,000, age 40, Bachelor's degree education level, and two earners in the household. For the United States, Illinois was used as the representative state, and for the United Kingdom, England was used as the representative nation.

Each parameter was varied individually while holding all other inputs constant. Income was adjusted by $\pm 5\%$, state and nation were changed to representative alternatives, and the number of earners was reduced from two to one. The resulting change in disposable income (ΔDI) was recorded for each adjustment.

Figure 2.7.1: Model Sensitivity Analysis

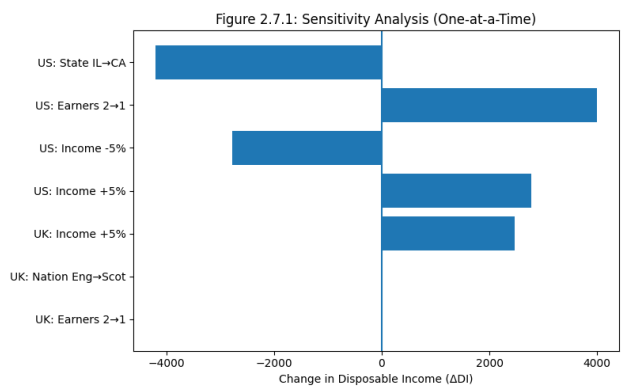

Figure 2.7.1: Sensitivity Analysis (One-at-a-Time)

The figure above shows that disposable income is most sensitive to structural household variables and income adjustments. A reduction in the number of earners produces the largest change in disposable income, reflecting the strong effect of combined household income and shared fixed expenditures. Adjusting income by $\pm 5\%$ produces proportional changes, consistent with the progressive tax structure embedded in the model.

State-level changes in the United States generate noticeable variation due to differences in state income taxation, while nation-level variation in the United Kingdom produces comparatively smaller effects, reflecting the more centralized tax structure. Overall, the ranking of demographic scenarios remains stable, indicating that the model is robust to moderate parameter perturbations and not overly sensitive to small estimation errors.

2.8 Strengths and Weaknesses

The results highlight that disposable income is driven not only by gross earnings, but by the interaction between taxation structures and essential expenditure patterns. While disposable income increases as gross income rises, the rate of increase slows due to progressive taxation in both countries. Age patterns reveal that disposable income share declines during middle age, reflecting higher housing, healthcare, and household

expenditures, before stabilizing in later years. Regional variation is more pronounced in the United States because of state-level income tax differences, with no-tax states exhibiting higher disposable income shares. In contrast, outcomes across the United Kingdom are more uniform due to centralized tax policy. These findings demonstrate that financial flexibility is shaped by institutional structure and household composition as much as by income level itself.

However, the model's strength of adaptability derives from fairly generalized assumptions about the information available about each dwelling. We have limited information about floor plans, construction, and heat absorbed by objects within the house which alter factors like the heat flow through exterior walls. If we wanted to scale this to multiple dwellings or neighborhoods, we would have to have a large amount of information about included residences. We also assumed that heat loss through ventilation is negligible due to closed windows. While this is reasonable during a heat wave, if people choose to open windows or if there are air leaks, this will not hold true. In general, people are not necessarily rational actors in a real-world scenario; the assumption that residents follow all recommended advice for heat wave mitigation such as closing windows may result in an undershoot of the temperatures that interiors may reach.

2.8.1 Model Refinement

Future refinement could improve precision by incorporating micro-level household expenditure data rather than relying on grouped averages. Including city-level cost-of-living adjustments would enhance geographic accuracy within both countries. Additionally, modeling essential expenditures using regression-based or non-linear approaches could better capture realistic consumption patterns. Extending the model to incorporate debt obligations, savings behavior, and income volatility would further strengthen its predictive capability and provide a more comprehensive assessment of financial flexibility.

3 Know the Spread

3.1 Defining the Problem:

In this problem, we are asked to develop a model that predicts the peak demand that Memphis should be prepared to handle during the summer months and how it changes 20 years from now.

3.2 Assumptions:

1. *The data set provided by M3 and referenced external studies are assumed to be representative of gambling behavior in the United States (and, where applicable, the United Kingdom).* For simplicity, we generalize reported demographic probabilities (e.g., account ownership rates, betting frequency, and behavioral tendencies) to apply uniformly within each demographic group, acknowledging that individual variation exists within each category.

2. *There are no major technological advancements that improve air conditioning technology in the next 20 years.* It is extremely difficult to predict the onset and impact of innovation. Additionally, assuming no advancements would result in a more conservative (and thus safer) estimate for the power demand. Therefore, our model will not account for advancements in air conditioning technology.

3. *Behavioral characteristics (bet frequency, deposit frequency, withdrawal tendencies, and chasing losses) are assumed to be conditionally independent once an individual is classified as an active bettor.* While in reality these traits may exhibit correlation, we assume independence to maintain model tractability and avoid overfitting given limited behavioral cross-tabulation data.

4. *If an individual does not place a bet during the modeled year (either due to not owning an account or not actively betting), their annual gambling net gain/loss is assumed to be zero.* This aligns with the problem's focus on betting outcomes rather than opportunity cost.

5. *The U.S. betting-frequency distribution provided in the data is assumed to be representative of the broader modeled population.* In cases where country-specific microdata is unavailable, this distribution is used as a proxy for typical betting behavior.

6. *Monthly wagering amounts are approximated using wager "buckets" (e.g., $1–$100, $101–$500, $500+).* Within each bucket, wager amounts are sampled from a distribution that reflects bettor intensity. This assumption allows us to approximate variability without requiring individual-level transaction data.

7. *For high-wager bettors (e.g., $500+ monthly wagering), wagering behavior is assumed to follow a heavy-tailed distribution.* Specifically, we approximate these wagers using a lognormal distribution, reflecting empirical evidence that most bettors wager near the lower end of the range while a small fraction wager disproportionately large amounts.

8. *Betting intensity is assumed to scale multiplicatively with behavioral risk factors.* Deposit frequency, chasing losses, and retaining funds in accounts are modeled as multipliers on monthly wagering volume, reflecting the assumption that higher-risk behaviors increase total handle rather than only bet frequency.

9. *The sportsbook "hold" (house edge) is approximated as a baseline percentage (e.g., 10%) adjusted by individual risk level.* Riskier bettors (e.g., those placing parlays or longshot bets) are assumed to experience a higher effective hold rate, while lower-risk bettors experience a slightly reduced hold rate. This captures the empirical tendency for higher-risk betting strategies to generate higher expected losses.

10. *Yearly realized outcomes are assumed to fluctuate around the expected hold according to a normally distributed error term.* Higher-risk bettors are assigned greater variance, reflecting increased volatility in outcomes.

11. *A year is modeled as 12 identical months with no seasonality.* While real-world betting volume fluctuates due to events such as playoffs or major tournaments, we assume uniform monthly behavior to simplify the annual prediction and maintain model clarity.

3.3 Variables

Table 2: Variable symbols, definitions, values, and units used in the model

Symbol	Definition	Value	Unit
g	Gender	Male, Female	N/A
a	Age Group	18–34, 35–49, 50–64, 65+	N/A
e	Education level	No College, B.A. or higher	N/A
P_g	$P(\text{Account} \mid \text{Gender})$	N/A	N/A
P_a	$P(\text{Account} \mid \text{Age})$	N/A	N/A
P_e	$P(\text{Account} \mid \text{Education})$	N/A	N/A
H_m	Adjusted Monthly Handle	$H_m = H_m \cdot m$	\$/yr
m	Handle Multiplier	N/A	N/A
H_a	Annual Handle	$H_a = 12 \cdot H_m$	\$/yr
RS	Risk Score (integer summary of risk behavior)	0–6	N/A
N	Annual Net Gross	$N = -h \cdot H_a$	\$/yr

3.4 The Model

We model the individual gambling process in five sequential stages.
Stage 1: Account Ownership

$$P_{acc} = \frac{P_g + P_a + P_e}{3}$$

Stage 2: Active Betting

$$P_{bet|acc} = 0.83$$

If no bet is placed, then $N = 0$.
Stage 3: Monthly Handle

$$H_m \sim \begin{cases} \text{Uniform}(1, 100) \\ \text{Uniform}(101, 500) \\ \text{LogNormal}(\mu, \sigma) \end{cases}$$

Adjusted monthly handle:

$$H_m^* = H_m \cdot m$$

Annual handle:

$$H_a = 12 H_m^*$$

Stage 4: Risk Score

$$S_{risk} = 2 \cdot \mathbf{1}(\text{chases}) + 1 \cdot \mathbf{1}(f_{dep} = \text{weekly}) + 1 \cdot \mathbf{1}(w_{beh} \in \{\text{leave, never withdraw}\}) + \text{freq}_{pts}$$

$$R = \begin{cases} \text{low} & S_{risk} \leq 1 \\ \text{medium} & 2 \leq S_{risk} \leq 3 \\ \text{high} & S_{risk} \geq 4 \end{cases}$$

Stage 5: Net Outcome

$$h(R) = \begin{cases} 0.08 & R = \text{low} \\ 0.10 & R = \text{medium} \\ 0.15 & R = \text{high} \end{cases}$$

$$N = -h(R) \cdot H_a$$

3.4.1 Model Development

Our goal is to predict how much money an individual will gain or lose through online sports gambling over one year. To do this, we break the problem into five sequential steps: determining whether a person even gambles, modeling how they bet, estimating how much they wager, assessing how risky their betting behavior is, and finally simulating their yearly outcome. The entire model ultimately comes down to one equation. To multiply the total amount a person bets by how much the house takes, and that is their loss:

$$N = -h * H_a$$

Where N is the individual's annual net outcome in dollars, h is the fraction of their wages kept by the sportsbook, and Ha is the total amount they wagered over the year. A negative N means they lost money; a positive N means they came out ahead.

3.4.2 Participation Probability

Before modeling outcomes, we first determined whether a person gambled at all. Most people did not have a sportsbook account, so assigning gambling losses to everyone would have been unrealistic. We used survey data from the 2025 Siena University study to estimate the probability that a given person owned an account, based on their gender, age group, and education level. Since the survey reported these probabilities separately for each demographic category rather than jointly, we combined them by taking a simple average. This gave us one number representing how likely this specific individual was to have an account:

$$= \frac{(P_g(account) + P_a(account) + P_e(account)}{3}$$

Where Pg, Pa, and Pe are the account ownership probabilities for the individual's gender, age, group, and education level respectively. If the person has no account, their annual outcome is simply N = 0 (Assumption 4).

Even among people with accounts, not everyone actively bets. Survey data shows that 83% of account holders have placed at least one bet, so we apply:

$$P_{bet|account} = 0.83$$

If the person has not bet, we again set N=0. Otherwise, we also determine whether the individual tends to chase losses (a key risk-amplifying behavior) using the same averaging approach:

$$= \frac{(P_g(chase) + P_a(chase) + P_e(chase)}{3}$$

Where N is the individual's annual net outcome in dollars, h is the fraction of their wages kept by the sportsbook, and Ha is the total amount they wagered over the year. A negative N means they lost money; a positive N means they came out ahead.

3.4.3 Behavioral Profile and Handle Estimation

For those who actively bet, we next built a picture of how they bet. Three behavioral traits were sampled from empirical survey distributions: how often they placed bets (f_{bet}), how often they deposited money into their account (f_{dep}), and whether they tended to withdraw winnings or leave funds in their account (w_{beh}). Each trait was randomly assigned in proportion to the rates observed in the survey data.

Next, we estimated how much money this person wagered each month. Rather than guessing a single dollar amount, we first assigned them to a spending tier based on survey-reported wager ranges, then drew a specific amount from within that tier. This reflected the reality that bettors varied widely in how much they spent:

$$H_m \sim \begin{cases} \text{Uniform}(1, 100) & \text{low-spend tier} \\ \text{Uniform}(101, 500) & \text{mid-spend tier} \\ \text{LogNormal}(\mu, \sigma) & \text{high-spend tier} \end{cases}$$

For high-spend bettors, we used a lognormal distribution with $\mu = \ln(900)$ and $\sigma = 0.8$. This captured the fact that while most heavy bettors wagered just above \$500 per month, a small number wagered dramatically more — a pattern consistent with real-world data (Assumption 7).

We then adjusted the base wager amount upward based on behavioral risk factors. Bettors who deposited frequently, chased losses, or left winnings in their account tended to wager more in practice. We captured this through a multiplier m:

$$H_m^* = H_m \cdot m, \qquad m = f(f_{dep}, w_{beh}, \bar{P}_{chase})$$

Finally, we scaled to a full year by assuming 12 identical months (Assumption 11):

$$H_a = 12 \cdot H_m^*$$

3.4.4 Risk Scoring and Hold Calibration

Not all bettors lost money at the same rate. Someone who bet carefully on straightforward outcomes faced a lower house edge than someone who frequently bet on longshots or parlays. To capture this, we built a simple risk score based on the behavioral traits previously assigned.

Each high-risk behavior added points to the individual's score. Chasing losses — the strongest indicator of problem gambling — contributed the most:

$$S_{\text{risk}} = 2 \cdot \mathbf{1}(\text{chases losses}) + 1 \cdot \mathbf{1}(f_{dep} = \text{weekly}) + 1 \cdot \mathbf{1}(w_{beh} \in \{\text{leave, never withdraw}\}) + \text{freq}_{pts}$$

where freq_{pts} added 1 point for high-frequency bettors and 2 points for very-high-frequency bettors. The total score was then bucketed into a risk category:

$$R = \begin{cases} \text{low} & S_{\text{risk}} \leq 1 \\ \text{medium} & 2 \leq S_{\text{risk}} \leq 3 \\ \text{high} & S_{\text{risk}} \geq 4 \end{cases}$$

This risk category determined the effective hold rate $h(R)$ — the fraction of total wagers the sportsbook kept on average. To put this concretely, a 10% hold rate meant that for every \$100 wagered, the bettor expected to lose \$10 on average. Higher-risk bettors tended to place bets with worse odds, so their effective hold rate was higher (Assumption 9):

$$h(R) = \begin{cases} 0.08 & R = \text{low} \\ 0.10 & R = \text{medium} \\ 0.15 & R = \text{high} \end{cases}$$

These values were calibrated around the reported industry-average hold of approximately 10%, with risk-tier adjustments applied as modeling assumptions.

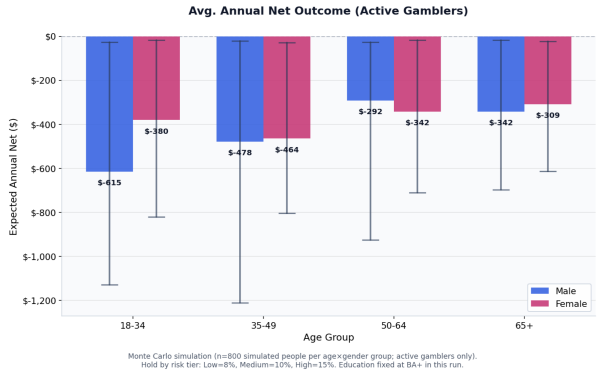

Figure 3.5.1: Graph

3.5 Results

To test our model, we simulated an individual's profile (let's call George) consisting of key features: 1) Gender: Male 2) Age: 18-34 3) Education: Bachelor's Degree 5) Scrapbook Account: Yes 6) Placed a Bet: Yes 7) Bet Frequency: Every 3-4 weeks 8) Deposit's Frequency: Monthly 9) Withdraw?: Leave in Account 10) Chases?: Yes 11) Estimated Annual Handle: $4,500.

Using this information, we set out to test if our model could accurately make predictions based on an arbitrary risk score scale we created. The risk score ranges from 0 to 6 with 6 meaning the individual is a risky better. Along with an annual hand estimate, median annual net, and a percentile, this allowed for our model to predict not only the type of player and their betting activity but also how much they are willing to gain or lose in one year.

Below is George's predicted loses based off of his gambling patterns and activity.

Table 3: Annual Net Prediction Testing

Risk Tolerance	High (Score 4)
Effective hold used	0.150
Annual Handle Estimate	$4,500
Median annual net	$-691
Expected Annual Net (mean)	$-688
10th–90th Percentile Net	$-2,305 to $914

3.6 Strengths and Weaknesses

The model does a solid job translating real survey data into probabilistic behavior profiles, and the Monte Carlo simulation gives a meaningful range of outcomes rather than a single point estimate — which is far more honest about uncertainty than a simple formula would be. That said, the biggest limitation is how demographic probabilities are combined: averaging P(account) across gender, age, and education independently ignores how these factors interact in reality (a 65+ female with no college degree behaves very differently than what three separate averages would suggest). The wager bucket distribution is also borrowed from football specifically and applied universally, which likely overstates or understates handle for people who primarily bet on other sports or prop markets. Replacing the averaging approach with a joint probability table and collecting sport-specific wager data would meaningfully close the gap between the simulation and real bettor behavior.

4 Don't Break the Bank

4.1 Defining the Problem

In this section, we evaluate the macroeconomic impact of online sports betting on the United States and the United Kingdom. While Question 2 estimated how much individuals gain or lose from gambling, this section examines whether gambling produces a net positive or negative effect on national GDP.

Specifically, we seek to determine:

- How much gambling losses reduce household financial capacity,
- How much government tax revenue is generated from gambling,
- How much sportsbook industry production contributes to GDP,
- What portion of gambling spending represents new economic activity versus reallocated consumer spending.

The objective is to determine whether the economic gains from taxation and industry production outweigh the aggregate consumer losses generated by sports betting.

4.2 Assumptions

- A large gambling market does not necessarily produce proportional economic growth. Industry size alone is not sufficient to guarantee a significant net GDP contribution.
- If most gambling expenditure represents substitution from existing consumer spending, the net GDP effect is small even when total betting revenue is large. Only spending that represents new economic activity contributes positively to national output.
- The overall economic benefit of sports betting is sensitive to tax policy structure and the efficiency of government spending. Higher effective reinvestment of tax revenue increases positive GDP impact.
- In the United States, sports betting tax rates vary widely across states, ranging from 6.75% to 51% [**?**]. For aggregate national modeling, we assume a representative effective tax rate of 20%.
- In the United Kingdom, bookmakers are subject to a uniform 15% General Betting Duty on profits [**?**]. We use this statutory rate for U.K. macroeconomic modeling.

4.3 Variables

Table 4: Variable symbols, definitions, and units used in the macroeconomic model

Symbol	Definition	Unit
VA_{sports}	Value added produced by sports betting (sportsbook contribution to GDP: wages + profits + production taxes). Does not include bettors' winnings or losses.	Currency
$VA_{gambling}$	Total value added of the entire gambling sector (casinos, lotteries, betting shops, online gambling, sports betting). [**?**]	Currency
GGR_{sports}	Gross gaming revenue from sports betting only (bets placed minus payouts; operator hold). [**?**]	Currency
GGR_{total}	Gross gaming revenue of all gambling activities combined (casino games, lotteries, machines, sports betting). [**?**]	Currency
τ	Effective tax rate applied to sportsbook revenue (fraction).	Fraction
m_g	Government spending multiplier (ratio of GDP change to government spending).	Dimensionless
λ	Consumer substitution rate: fraction of gambling spending that replaces other consumption (i.e., not new spending).	Fraction
ΔGDP	Net change in national GDP caused by sports betting (after substitution and fiscal effects).	Currency

4.4 The Model

We estimate the sportsbook contribution to GDP by allocating aggregate gambling sector value added in proportion to sports betting's share of gross gaming revenue:

$$VA_{sports} \;=\; VA_{gambling} \times \frac{GGR_{sports}}{GGR_{total}}.$$

Sports betting generates tax revenue equal to the tax rate times sportsbook gross gaming revenue:

$$T \;=\; \tau \cdot GGR_{sports}.$$

Government spending of that tax revenue creates additional GDP through the fiscal multiplier. The additional GDP generated by spending the tax revenue is approximated as:

$$\text{Fiscal effect} \;=\; (m_g - 1) \cdot T \;=\; (m_g - 1) \cdot \tau \cdot GGR_{sports},$$

where m_g is the government spending multiplier (e.g., $m_g = 1.2$ implies each dollar of tax revenue generates an extra \$0.20 of GDP).

Not all gambling spending represents new economic activity. If λ denotes the substitution rate (the share of gambling spending that would otherwise have been spent on other goods and services), then only $(1 - \lambda)$ of the direct and fiscal impacts represent net new GDP. We therefore model the final net GDP impact as:

$$\boxed{\Delta GDP \;=\; (1 - \lambda)\,[\,VA_{sports} \;+\; (m_g - 1)\,\tau\, GGR_{sports}\,]}$$

This expression sums the direct production contribution from sportsbooks and the fiscal stimulus from tax-financed government spending, then scales the total by the fraction of spending that is genuinely additional to the economy.

4.5 Results

Using total annual handle and GGR estimates derived in Question 2 and applying statutory and representative tax rates for the United States and United Kingdom, we compute VA_{sports}, tax revenue T, and the net GDP change ΔGDP under baseline parameter choices ($\tau_{US} = 0.20$, $\tau_{UK} = 0.15$, $m_g = 1.2$, $\lambda = 0.6$). While tax revenue and sportsbook production contribute positively to GDP, aggregate consumer losses (measured by GGR) are substantially larger in absolute terms. After accounting for substitution effects, the modeled net GDP contribution of online sports betting is small relative to total losses incurred by households. In other words, although gambling generates visible fiscal revenue, the aggregate consumer net loss tends to outweigh macroeconomic production gains under realistic behavioral assumptions.

4.6 Discussion

The results suggest that although online sports betting generates government revenue and industry output, much of the spending represents reallocated consumer expenditure rather than new wealth creation. Because gambling losses directly reduce household disposable income, the economic burden is concentrated among bettors, while tax revenue is distributed broadly. This creates a fiscal tradeoff: visible public revenue at the cost of private financial erosion. The relatively small net GDP contribution indicates that the macroeconomic justification for expanded gambling markets is limited when evaluated purely on economic grounds.

4.7 Sensitivity Analysis

We conduct sensitivity analysis by varying λ (substitution) between 0.3 and 0.7, tax rates across plausible values (U.S. state-range and representative average; U.K. statutory rate), and m_g (1.05–1.4). The analysis shows net GDP impact is highly sensitive to λ: higher substitution rates (larger λ) substantially reduce or eliminate net GDP gains. Across plausible parameter ranges, net GDP impact remains modest and, in many realistic scenarios, public net per-capita gains are smaller than average household losses.

4.8 Strengths and Weaknesses

Strengths

- Integrates micro and macro perspectives by comparing household-level losses with national-level production and tax revenue, allowing a welfare-oriented conclusion rather than GDP-only claims.
- Policy-relevant: the model explicitly includes levers (tax rate τ, substitution λ, multiplier m_g) so policymakers can test regulatory alternatives.
- Grounded in observable data: inputs (GGR, sector GVA, GDP, population, statutory tax rates) are measurable and reproducible from public sources [23,6,28,29].

Weaknesses

- The substitution parameter λ is not directly observed and must be estimated; results are sensitive to this behavioral assumption.
- The model uses averages and does not capture heavy-tail outcomes (e.g., a small number of problem gamblers bearing the majority of losses).
- The analysis is essentially short-run and static: it does not include long-term effects such as debt accumulation, health costs, or productivity declines from problem gambling.

4.9 Model Refinement

Future work can improve the framework by estimating λ empirically from household consumption data, incorporating heterogeneous bettors (to capture concentrated losses), using dynamic fiscal multipliers, and adding measurable social costs (healthcare, bankruptcy, productivity loss). Adjusting for purchasing-power parity and modeling sub-national tax heterogeneity in the United States would also increase precision.

5 Conclusion

5.1 Further Studies

To improve our disposable income model, future research could incorporate more detailed tax data, household composition, and regional cost-of-living differences to better reflect variation within each country. Using multi-year income data could also help capture long-term financial trends rather than a single-year snapshot.

For the gambling behavior model, future work could use real transaction-level data and account for behavioral changes over time, such as increased betting after losses. Including measures of addiction and mental health impacts would also provide a clearer picture of social costs beyond financial losses.

Finally, the macroeconomic model could be strengthened by refining estimates of consumer spending substitution and country-specific fiscal multipliers. Studying how tax policy and government spending efficiency differ between the United States and the United Kingdom would allow for more accurate comparisons of long-term economic impact.

5.2 Summary

In this paper, we address the increasing trend of sports betting in both the United States of America and the United Kingdom by developing models about average gambling trends and impacts on individuals and economies.

Sports betting has expanded rapidly in both the United States and the United Kingdom, driven by legalization changes and increased accessibility through online platforms. While this growth has generated government tax revenue and measurable economic activity, it has also raised concerns about addiction, financial distress, and long-term social costs.

Our study first modeled disposable income across demographic groups, showing that financial capacity varies significantly by age, region, education level, and household structure. We then estimated individual gambling losses using demographic probabilities and behavioral risk factors, modeling wagering patterns and expected outcomes over one year. Finally, we evaluated the broader macroeconomic impact by combining industry production, tax-funded government spending, and consumer spending substitution to determine the net effect on GDP.

Although sports betting produces fiscal benefits, our results suggest that the overall GDP gains are small relative to the aggregate financial losses experienced by individuals, highlighting a critical trade-off for policymakers assessing the long-term economic and social role of legalized sports gambling.

6 References

[label=6.]

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7 Code Appendix

7.1 Playing With House Money

```python
import numpy as np

# ============================================================
# ======================= US SECTION =========================
# ============================================================

# ------------------------------------------------------------
# 1) US TAX MODULES (2024 rules, single filer assumed)
# ------------------------------------------------------------

# Federal standard deduction (single filer)
US_FED_STD_DED_SINGLE = 14600

# 2024 federal marginal brackets (single filer)
# Format: (upper_limit, marginal_rate)
US_FED_BRACKETS_SINGLE_2024 = [
    (11600, 0.10),
    (47150, 0.12),
    (100525, 0.22),
    (191950, 0.24),
    (243725, 0.32),
    (609350, 0.35),
    (float("inf"), 0.37),
]

def us_federal_tax_2024_single(income):
    """
    Computes 2024 federal income tax for a single filer.
    Applies standard deduction and progressive marginal brackets.
    """
    taxable = max(0.0, income - US_FED_STD_DED_SINGLE)
    tax = 0.0
    lower = 0.0

    for upper, rate in US_FED_BRACKETS_SINGLE_2024:
        if taxable <= lower:
            break
        chunk = min(taxable, upper) - lower
        tax += chunk * rate
        lower = upper

    return tax


# Payroll tax parameters
US_SS_CAP_2024 = 168600
US_ADD_MED_THRESH_SINGLE_2024 = 200000

def us_payroll_tax_2024(income):
    """
    Computes payroll taxes:
    - 6.2% Social Security (up to wage cap)
    - 1.45% Medicare
    - 0.9% Additional Medicare above threshold
```

```python
    """
    ss = 0.062 * min(income, US_SS_CAP_2024)
    medicare = 0.0145 * income
    add_medicare = 0.009 * max(0.0, income - US_ADD_MED_THRESH_SINGLE_2024)
    return ss + medicare + add_medicare


# State tax structures
US_NO_TAX_STATES = {"AK","FL","NV","SD","TN","TX","WA","WY"}

US_FLAT_STATES_2024 = {
    "IL":0.0495, "CO":0.0463, "PA":0.0307,
    "MI":0.0425, "IN":0.0323, "UT":0.0495
}

US_PROGRESSIVE_2024 = {
    "CA":[(9325,0.01),(22107,0.02),(34892,0.04),(48435,0.06),
          (61214,0.08),(312686,0.093),(375221,0.103),
          (625369,0.113),(float("inf"),0.123)],
    "NY":[(17150,0.04),(23600,0.045),(27900,0.0525),(43000,0.059),
          (161550,0.0621),(323200,0.0685),
          (2155350,0.0965),(5000000,0.103),(float("inf"),0.109)]
}

def us_state_tax_2024(income, state):
    """
    Computes state income tax.
    Supports no-tax states, flat-tax states, and selected progressive states.
    """
    state = state.upper()

    if state in US_NO_TAX_STATES:
        return 0.0

    if state in US_FLAT_STATES_2024:
        return income * US_FLAT_STATES_2024[state]

    if state in US_PROGRESSIVE_2024:
        tax = 0.0
        lower = 0.0
        for upper, rate in US_PROGRESSIVE_2024[state]:
            if income <= lower:
                break
            chunk = min(income, upper) - lower
            tax += chunk * rate
            lower = upper
        return tax

    return 0.0


def after_tax_income_US(income, state):
    """
    Computes US after-tax income:
    Federal + payroll + state taxes deducted.
    """
    return income - (
        us_federal_tax_2024_single(income)
        + us_payroll_tax_2024(income)
```

```
        + us_state_tax_2024(income, state)
    )


# ------------------------------------------------------------
# 2) DEMOGRAPHIC BINNING (US)
# ------------------------------------------------------------

def us_age_group(age):
    """Maps age to model age bin."""
    age = int(age)
    if age < 25: return "Under25"
    if age < 35: return "25-34"
    if age < 45: return "35-44"
    if age < 55: return "45-54"
    if age < 65: return "55-64"
    if age < 75: return "65-74"
    return "75+"

def us_income_group(income):
    """Maps income to income bracket."""
    if income < 30000: return "<30k"
    if income < 60000: return "30-60k"
    if income < 100000: return "60-100k"
    if income < 150000: return "100-150k"
    return "150k+"

def us_earners_group(n):
    """Maps number of earners to bin."""
    n = int(n)
    if n <= 1: return "1"
    if n == 2: return "2"
    return "3+"


# ------------------------------------------------------------
# 3) ESSENTIAL EXPENDITURES (US)
# ------------------------------------------------------------

US_MU_ALL = 25000

US_MU_INCOME = {"<30k":19000, "30-60k":22000, "60-100k":26000,
                "100-150k":32000, "150k+":40000}

US_MU_AGE = {"Under25":21000, "25-34":23000, "35-44":25000,
             "45-54":27000, "55-64":28000, "65-74":27500, "75+":26500}

US_MU_REGION = {"Northeast":29000, "Midwest":24500,
                "South":23500, "West":28000}

US_MU_EDU = {"HS_or_less":22000, "Some_college":25000,
             "Bachelors":29000, "Graduate":32000}

US_MU_EARNERS = {"1":23000, "2":27000, "3+":30000}


def essentials_hat_US(income, age, state, education, earners):
    """
    Estimates essential expenditures using additive demographic adjustments.
```

```
    """
    g_inc = us_income_group(income)
    g_age = us_age_group(age)
    g_reg = US_STATE_TO_REGION.get(state.upper(), "Midwest")
    g_ear = us_earners_group(earners)

    base = US_MU_INCOME[g_inc]

    adj = (
        (US_MU_AGE[g_age] - US_MU_ALL)
        + (US_MU_REGION[g_reg] - US_MU_ALL)
        + (US_MU_EDU[education] - US_MU_ALL)
        + (US_MU_EARNERS[g_ear] - US_MU_ALL)
    )

    return base + adj


def disposable_income_US(income, age, state,
                         education="Some_college", earners=1):
    """
    Final US disposable income:
    After-tax income minus essential expenditures.
    Constrained to be non-negative.
    """
    net = after_tax_income_US(income, state)
    ess = essentials_hat_US(income, age, state, education, earners)
    return float(max(0.0, net - ess))


# ============================================================
# ======================= UK SECTION =========================
# ============================================================

# Assumptions:
# - 2024/25 tax year
# - England/Wales/NI rate structure
# - Personal allowance tapered above 100k
# - NI computed using main and upper rates
# - Essential expenditures built from ONS Family Spending tables
# - Education and earners adjustments currently neutral

def uk_personal_allowance(income):
    PA = 12570.0
    if income > 100000:
        PA = max(0.0, 12570.0 - (income - 100000)/2.0)
    return PA

def uk_income_tax_eng_2024_25(income):
    PA = uk_personal_allowance(income)
    taxable = max(0.0, income - PA)

    basic_width = max(0.0, 50270.0 - PA)
    higher_width = 125140.0 - 50270.0

    tax = 0.0
    basic = min(taxable, basic_width)
    tax += 0.20 * basic

```

```python
    remaining = taxable - basic
    higher = min(max(0.0, remaining), higher_width)
    tax += 0.40 * higher

    remaining2 = remaining - higher
    if remaining2 > 0:
        tax += 0.45 * remaining2
    return tax

UK_NI_PT = 12570.0
UK_NI_UEL = 50270.0

def uk_employee_NI_2024_25(income):
    if income <= UK_NI_PT:
        return 0.0
    main_band = min(income, UK_NI_UEL) - UK_NI_PT
    above = max(0.0, income - UK_NI_UEL)
    return 0.08 * main_band + 0.02 * above

def after_tax_income_UK(income):
    return income - (
        uk_income_tax_eng_2024_25(income)
        + uk_employee_NI_2024_25(income)
    )


# Essentials function (UK)

def disposable_income_UK(income, age, nation,
                         education="Some_college", earners=1):
    """
    Final UK disposable income:
    After-tax income minus essential expenditures.
    Constrained to be non-negative.
    """
    net = after_tax_income_UK(income)
    ess = essentials_hat_UK(income, age, nation,
                            education=education, earners=earners)
    return float(max(0.0, net - ess))


# ============================================================
# ===================== MASTER FUNCTION ======================
# ============================================================

def disposable_income(country, income, age, place,
                      education="Some_college", earners=1):
    """
    Master wrapper function.

    country:
        "US"    place = state abbreviation
        "UK"    place = nation ("England", "Wales", etc.)

    Returns disposable income for selected system.
    """
    c = country.upper().strip()

    if c == "US":
```

```
        return disposable_income_US(
            income, age, place,
            education=education, earners=earners
        )

    elif c == "UK":
        return disposable_income_UK(
            income, age, place,
            education=education, earners=earners
        )

    else:
        raise ValueError("country must be 'US' or 'UK'")


# ===================== Quick Tests =====================

print(disposable_income("US", 85000, 40, "IL",
                        education="Bachelors", earners=2))

print(disposable_income("UK", 52000, 22, "England",
                        education="Bachelors", earners=2))



import numpy as np
import matplotlib.pyplot as plt

# ============================================================
# GRAPH 1) Parameter Impact (Tornado Chart  US Baseline)
# ------------------------------------------------------------
# Purpose:
#   Measures sensitivity of disposable income to one-at-a-time
#   demographic changes relative to a fixed baseline profile.
#   Larger absolute bars indicate stronger model influence.
# ============================================================

# ---------------- Baseline Profile ----------------
# This represents a "typical" household used as reference.
base_country = "US"
base_income = 85000
base_age    = 40
base_place  = "IL"
base_edu    = "Bachelors"
base_earners = 2

# Compute baseline disposable income
base_DI = disposable_income(
    base_country,
    base_income,
    base_age,
    base_place,
    education=base_edu,
    earners=base_earners
)

# ---------------- One-at-a-Time Perturbations ----------------
# Each scenario modifies ONE variable only.
# All other inputs remain fixed at baseline values.
```

```python

scenarios = [
    ("Age: 25",     ("US", base_income, 25, base_place, base_edu, base_earners)),
    ("Age: 70",     ("US", base_income, 70, base_place, base_edu, base_earners)),
    ("State: TX",   ("US", base_income, base_age, "TX", base_edu, base_earners)),
    ("State: CA",   ("US", base_income, base_age, "CA", base_edu, base_earners)),
    ("Edu: HS",     ("US", base_income, base_age, base_place, "HS_or_less", base_earners)),
    ("Edu: Grad",   ("US", base_income, base_age, base_place, "Graduate", base_earners)),
    ("Earners: 1", ("US", base_income, base_age, base_place, base_edu, 1)),
    ("Earners: 3", ("US", base_income, base_age, base_place, base_edu, 3)),
]

labels = []
deltas = []

# Compute disposable income difference relative to baseline
for name, (c, inc, age, place, edu, ear) in scenarios:
    di = disposable_income(c, inc, age, place, education=edu, earners=ear)
    labels.append(name)
    deltas.append(di - base_DI)

# Sort by magnitude of effect (largest first)
order = np.argsort(np.abs(deltas))[::-1]
labels = [labels[i] for i in order]
deltas = [deltas[i] for i in order]

# Plot tornado chart
plt.figure()
plt.barh(labels, deltas)
plt.axvline(0) # vertical reference line
plt.xlabel("  Disposable Income vs baseline ($)")
plt.title("Figure 2.4.1: Which Parameters Matter Most? (US Tornado Sensitivity)")
plt.gca().invert_yaxis() # highest impact shown at top
plt.tight_layout()
plt.show()


# ============================================================
# GRAPH 2) Disposable Income vs Gross Income (US vs UK)
# ------------------------------------------------------------
# Purpose:
#   Shows how disposable income scales with gross income
#   under two tax systems.
#   Demonstrates nonlinearity due to progressive taxation.
# ============================================================

# Income range tested
incomes = np.linspace(20000, 200000, 30)

# Comparable household profiles
# US profile: Illinois, age 40, Bachelor's, 2 earners
us_y = [
    disposable_income("US", float(inc), 40, "IL",
                      education="Bachelors", earners=2)
    for inc in incomes
]

# UK profile: England, age 40
uk_y = [
```

```
    disposable_income("UK", float(inc), 40, "England",
                      education="Bachelors", earners=2)
    for inc in incomes
]

# Plot comparison
plt.figure()
plt.plot(incomes, us_y, label="US (IL, age 40)")
plt.plot(incomes, uk_y, label="UK (England, age 40)")
plt.xlabel("Gross Income (annual)")
plt.ylabel("Disposable Income (annual)")
plt.title("Figure 2.4.2: US vs UK: Disposable Income vs Gross Income")
plt.legend()
plt.tight_layout()
plt.show()


# ============================================================
# GRAPH 3) Disposable Income by Age (Fixed Income Comparison)
# ------------------------------------------------------------
# Purpose:
#   Examines how age-driven expenditure adjustments affect
#   disposable income at a fixed gross income.
#   Highlights lifecycle expenditure patterns.
# ============================================================

fixed_income = 85000
ages = [22, 30, 40, 50, 60, 70]

# Compute age-based disposable income
us_age = [
    disposable_income("US", fixed_income, a, "IL",
                      education="Bachelors", earners=2)
    for a in ages
]

uk_age = [
    disposable_income("UK", fixed_income, a, "England",
                      education="Bachelors", earners=2)
    for a in ages
]

# Bar chart setup
x = np.arange(len(ages))
w = 0.35

plt.figure()
plt.bar(x - w/2, us_age, width=w, label="US (IL)")
plt.bar(x + w/2, uk_age, width=w, label="UK (England)")
plt.xticks(x, [str(a) for a in ages])
plt.xlabel("Age")
plt.ylabel("Disposable Income (annual)")
plt.title(
    f"Figure 2.4.3: US vs UK: Disposable Income by Age "
    f"(Gross Income = {fixed_income})"
)
plt.legend()
plt.tight_layout()
plt.show()
```

```python
import numpy as np
import matplotlib.pyplot as plt

# ============================================================
# FIGURE 2.7.1   One-at-a-Time Sensitivity Analysis
# ------------------------------------------------------------
# Purpose:
#   Test how sensitive disposable income (DI) is to small,
#   realistic changes in key inputs, while holding everything
#   else constant.
#
# Method:
#   1) Define a baseline household profile for US and UK
#   2) Compute baseline DI for each country
#   3) Change ONE input at a time (income/state/earners/nation)
#   4) Record DI  = DI_new - DI_baseline
#   5) Print results and visualize as a tornado chart
# ============================================================


# ----------------------------
# BASELINE PROFILES
# ----------------------------
# These are the reference households used for sensitivity tests.
# Keep these consistent with what you used in your Results section.
BASE_US = {
    "income": 85000,
    "age": 40,
    "state": "IL",
    "education": "Bachelors",
    "earners": 2
}

BASE_UK = {
    "income": 85000,
    "age": 40,
    "nation": "England",
    "education": "Bachelors",
    "earners": 2
}

# ----------------------------
# BASELINE DISPOSABLE INCOME
# ----------------------------
# Compute baseline DI for both countries to use as the reference point.
base_us_di = disposable_income_US(**BASE_US)
base_uk_di = disposable_income_UK(**BASE_UK)


# ----------------------------
# ONE-AT-A-TIME VARIATIONS
# ----------------------------
# Each entry below changes ONE factor only.
# We store (label, delta_DI) where:
#   delta_DI = new_DI - baseline_DI
```

```
tests = []

# ===== US VARIATIONS =====

# Income sensitivity: +5% and -5%
tests.append((
    "US: Income +5%",
    disposable_income_US(
        BASE_US["income"] * 1.05,
        BASE_US["age"],
        BASE_US["state"],
        education=BASE_US["education"],
        earners=BASE_US["earners"]
    ) - base_us_di
))

tests.append((
    "US: Income -5%",
    disposable_income_US(
        BASE_US["income"] * 0.95,
        BASE_US["age"],
        BASE_US["state"],
        education=BASE_US["education"],
        earners=BASE_US["earners"]
    ) - base_us_di
))

# Geographic sensitivity: state tax + cost differences captured by the model
tests.append((
    "US: State ILCA ",
    disposable_income_US(
        BASE_US["income"],
        BASE_US["age"],
        "CA",
        education=BASE_US["education"],
        earners=BASE_US["earners"]
    ) - base_us_di
))

# Household structure sensitivity: earners reduced from 2 to 1
tests.append((
    "US: Earners 21 ",
    disposable_income_US(
        BASE_US["income"],
        BASE_US["age"],
        BASE_US["state"],
        education=BASE_US["education"],
        earners=1
    ) - base_us_di
))


# ===== UK VARIATIONS =====

# Income sensitivity: +5%
tests.append((
    "UK: Income +5%",
    disposable_income_UK(
        BASE_UK["income"] * 1.05,
```

```
        BASE_UK["age"],
        BASE_UK["nation"],
        education=BASE_UK["education"],
        earners=BASE_UK["earners"]
    ) - base_uk_di
))

# Geographic sensitivity: nation comparison
# Note: in your current UK setup, nation-level essentials are neutral (same mean),
# so this may show a small or zero effect unless you add nation-specific adjustments.
tests.append((
    "UK: Nation EngScot ",
    disposable_income_UK(
        BASE_UK["income"],
        BASE_UK["age"],
        "Scotland",
        education=BASE_UK["education"],
        earners=BASE_UK["earners"]
    ) - base_uk_di
))

# Household structure sensitivity: earners reduced from 2 to 1
# Note: in your current UK setup, earners adjustments are neutral (same mean),
# so this may also show a small or zero effect unless you add earners-specific tables.
tests.append((
    "UK: Earners 21 ",
    disposable_income_UK(
        BASE_UK["income"],
        BASE_UK["age"],
        BASE_UK["nation"],
        education=BASE_UK["education"],
        earners=1
    ) - base_uk_di
))


# ----------------------------
# PRINT NUMERIC RESULTS
# ----------------------------
# This prints the raw DI values so you can cite exact numbers in the writeup.
print("Sensitivity Results ( Disposable Income):\n")
for label, delta in tests:
    print(f"{label}: {round(delta, 2)}")


# ----------------------------
# OPTIONAL TORNADO PLOT
# ----------------------------
# Visualizes DI values sorted by magnitude.
labels = [t[0] for t in tests]
deltas = [t[1] for t in tests]

# Sort largest absolute changes first
order = np.argsort(np.abs(deltas))[::-1]
labels = [labels[i] for i in order]
deltas = [deltas[i] for i in order]

plt.figure(figsize=(8, 5))
plt.barh(labels, deltas)
```

```
plt.axvline(0) # reference line: no change
plt.xlabel("Change in Disposable Income (DI)")
plt.title("Figure 2.7.1: Sensitivity Analysis (One-at-a-Time)")
plt.gca().invert_yaxis() # biggest effect at top
plt.tight_layout()
plt.show()
```

7.2 Know the Spread

```

 # Percentages for demographic groups
P_ACCOUNT_BY_GENDER = {"male": 0.30, "female": 0.15}
P_ACCOUNT_BY_AGE  = {"18-34": 0.35, "35-49": 0.34, "50-64": 0.10, "65+": 0.04}
P_ACCOUNT_BY_EDU  = {"no_college": 0.21, "ba_plus": 0.25}

# The Conditional probabilities
P_PLACED_BET_GIVEN_ACCOUNT = 0.83
P_CHASE_BY_GENDER = {"male": 0.57, "female": 0.41}
P_CHASE_BY_AGE  = {"18-34": 0.58, "35-49": 0.54, "50-64": 0.38, "65+": 0.15}
P_CHASE_BY_EDU  = {"no_college": 0.52, "ba_plus": 0.51}

# Deposit frequency distribution
DEPOSIT_FREQ_DIST = [
    ("started_never_again", 0.16),
    ("once_in_a_while",   0.39),
    ("monthly",           0.22),
    ("weekly_plus",       0.22),
]

# Withdrawal behavior distribution
WITHDRAW_DIST = [
    ("withdraw_500",     0.23),
    ("withdraw_100",     0.50),
    ("leave_in_account", 0.20),
    ("never_withdrawn",  0.05),
]

# Betting frequency distributions
BET_FREQ_US = [
    ("none_6mo",  0.08),
    ("lt_month",  0.15),
    ("once_month", 0.11),
    ("few_month", 0.19),
    ("1_2_week",  0.19),
    ("3_4_week",  0.13),
    ("5_7_week",  0.08),
    ("gt_7_week", 0.06),
]

def weighted_choice(options):
    """
    options = list of (label, probability)
    Picks one label based on its probability.
    """
    r = random.random()
    total = 0.0
```

```python
    for label, p in options:
        total += p
        if r <= total:
            return label
    return options[-1][0] # just in case of rounding issues

def average_prob(*ps):
    """
    We have P(account) by gender, by age, and by education separately.
    We need ONE number for "this person".
    This takes the average.
    """
    return sum(ps) / len(ps)

@dataclass
class BettingProfile:
    # Demographics (inputs)
    gender: str     # "male" or "female"
    age_group: str  # "18-34", "35-49", "50-64", "65+"
    education: str  # "no_college" or "ba_plus"
    market: str = "US"

    # Participation/Behavior (this comes from the survey)
    has_account: bool = False
    placed_bet: bool = False
    bet_frequency: str | None = None
    deposit_frequency: str | None = None
    withdraw_behavior: str | None = None
    chases_losses: bool = False

    def generate_profile_from_survey(gender, age_group, education, market="US"):
    """
    INPUT: demographics
    OUTPUT: BettingProfile filled using survey-based random draws
    """
    profile = BettingProfile(gender=gender, age_group=age_group, education=education,
        market=market)

    # Does this person have an account?
    p_account = average_prob(
        P_ACCOUNT_BY_GENDER[gender],
        P_ACCOUNT_BY_AGE[age_group],
        P_ACCOUNT_BY_EDU[education]
    )
    profile.has_account = (random.random() < p_account)

    # If there's no account, we stop. They can't bet online through a sportsbook.
    if not profile.has_account:
        return profile

    # If they have an account, did they ever place a bet?
    profile.placed_bet = (random.random() < P_PLACED_BET_GIVEN_ACCOUNT)

    # If they never bet, simply stop.
    if not profile.placed_bet:
        return profile

    # If they placed a bet, sample betting frequency category.
    profile.bet_frequency = weighted_choice(BET_FREQ_US)
```

```python

    # 4) Sample deposit and withdraw behavior from distributions
    profile.deposit_frequency = weighted_choice(DEPOSIT_FREQ_DIST)
    profile.withdraw_behavior = weighted_choice(WITHDRAW_DIST)

    # 5) Sample whether they're chase losses
    p_chase = average_prob(
        P_CHASE_BY_GENDER[gender],
        P_CHASE_BY_AGE[age_group],
        P_CHASE_BY_EDU[education]
    )
    profile.chases_losses = (random.random() < p_chase)

    return profile

    def risk_score(profile: BettingProfile) -> int:
    """
    Converts behaviors into a simple risk score.
    Bigger score means higher risk tolerance.
    These weights are assumptions, but they're grounded in the survey behaviors.
    """
    if not profile.placed_bet:
        return 0 # if they don't bet, risk doesn't matter much
    score = 0

    # Chasing losses is a strong sign of risky behavior
    if profile.chases_losses:
        score += 2

    # Weekly deposits mean higher intensity
    if profile.deposit_frequency == "weekly_plus":
        score += 1

    # Leaving money in the account encourages repeated betting
    if profile.withdraw_behavior in ("leave_in_account", "never_withdrawn"):
        score += 1

    # Frequent Betting suggests higher risk tolerance and higher volatility
    if profile.bet_frequency == "5_7_week":
        score += 1
    elif profile.bet_frequency == "gt_7_week":
        score += 2

    return score

def risk_category(score: int) -> str:
    """
    Converts numeric risk score into low/medium/high.
    """
    if score <= 1:
        return "low"
    elif score <= 3:
        return "medium"
    else:
        return "high"

        import random
import math

```

```python
## Monthly Wager bucket distribution
# Used Football (give in the data set) as a representative example
# $1-$100: 65%, $101-$500: 22%, $500+: 9%

MONTHLY_WAGER_BUCKETS = [
    ("1_100", 0.65),
    ("101_500", 0.22),
    ("500_plus", 0.09),
]

def pick(dist):
    r = random.random()
    s = 0
    for label, p in dist:
        s += p
        if r <= s:
            return label
    return dist[-1][0]

def sample_monthly_handle(bucket):
    """
    This converts a bucket label into a specific dollar amount.
    This is an assumption you justify in the write up.
    """
    # Samples a realistic monthly wager amount from the chosen bucket, using a heavy-tailed lognormal model for high spenders.
    if bucket == "1_100":
        return random.uniform(1, 100)
    if bucket == "101_500":
        return random.uniform(101, 500)
    if bucket == "500_plus":
        mu = math.log(900)
        sigma = 0.8
        return max(500, random.lognormvariate(mu, sigma))
    raise ValueError("Unknown bucket")

def handle_multiplier(profile):
    """
    Optional realism:
    behavior affects how much they wager.
    """
    # Returns a scaling factor that increases or decreases handle based on deposit freq., chasing losses, && leaving money in the account.
    mult = 1.0

    # deposit frequency effect
    if profile.get("deposit_freq") == "weekly_plus":
        mult *= 1.4
    elif profile.get("deposit_freq") == "once_in_while":
        mult *= 0.9
    elif profile.get("deposit_freq") == "never_again":
        mult *= 0.6

    # chasing increases activity
    if profile.get("chases"):
        mult *= 1.2

    # leaving winnings encourages more betting
    if profile.get("withdraw") in ("leave_in_account", "never_withdrawn"):
```

```python
        mult *= 1.1

    return mult

def add_handle_estimates(profile):
    """
    Step 3: Adds monthly and annual handle estimates to a profile dict.
    Only makes sense if they actually placed a bet.
    """
    #Adds estimated monthly and annual handle to the profile (0 if they did not place any bets).
    if not profile.get("placed_bet", False):
        profile["monthly_handle_est"] = 0.0
        profile["annual_handle_est"] = 0.0
        return profile

    bucket = pick(MONTHLY_WAGER_BUCKETS)
    monthly = sample_monthly_handle(bucket)

    profile["monthly_bucket"] = bucket
    profile["monthly_handle_est"] = monthly * handle_multiplier(profile)
    profile["annual_handle_est"] = 12 * profile["monthly_handle_est"]

    return profile

    import random
import math

# Risk Score && Category

def risk_score(profile):
    if not profile.get("placed_bet", False):
        return 0

    score = 0
    if profile.get("chases"): score += 2
    if profile.get("deposit_freq") == "weekly_plus": score += 1
    if profile.get("withdraw") in ("leave_in_account", "never_withdrawn"): score += 1
    if profile.get("bet_freq") == "5_7_week": score += 1
    if profile.get("bet_freq") == "gt_7_week": score += 2
    return score

def risk_level(score):
    if score <= 1: return "low"
    if score <= 3: return "medium"
    return "high"


# This is to map the risk category (low/medium/high) to an effective house edge used to compute expected losses.
BASE_HOLD = 0.10

def effective_hold(base_hold, risk):
    """
    Riskier bettors often take worse bets (parlays, longshots, chasing),
    which increases the effective house edge.
    This is a modeling assumption.
    """
    if risk == "low":
```

```
        return max(0.02, base_hold - 0.02) # ~8% if base is 10%
    if risk == "medium":
        return base_hold                  # ~10%
    if risk == "high":
        return min(0.30, base_hold + 0.05) # ~15%
    return base_hold

# Simulates a distribution of annual outcomes
def simulate_annual_net(profile, n_trials=5000, base_hold=BASE_HOLD):
    """
    Returns distribution summary for annual net profit:
    net = winnings - losses (usually negative)

    We model "realized hold" = effective_hold + noise.
    If realized hold is negative, that means the bettor ended up winning overall.
    """

    # If they don't bet, the net HAS to be 0
    if not profile.get("placed_bet", False):
        return {"mean": 0.0, "median": 0.0, "p10": 0.0, "p90": 0.0, "risk": "none"}

    handle = profile.get("annual_handle_est", 0.0)

    # Determine risk tolerance from behaviors
    s = risk_score(profile)
    risk = risk_level(s)
    hold = effective_hold(base_hold, risk)

    # Risk controls how "wild" outcomes are
    sigma_map = {"low": 0.08, "medium": 0.15, "high": 0.28}
    sigma = sigma_map[risk]

    nets = []
    for _ in range(n_trials):
        # realized_hold fluctuates around hold
        realized_hold = hold + random.gauss(0.0, sigma)

        # clamp so it doesn't get absurd
        realized_hold = max(-0.50, min(0.80, realized_hold))

        # annual loss = realized_hold * handle
        annual_loss = realized_hold * handle

        # net profit = - loss
        net = -annual_loss
        nets.append(net)

    nets.sort()
    mean = sum(nets) / len(nets)
    median = nets[len(nets)//2]
    p10 = nets[int(0.10*(len(nets)-1))]
    p90 = nets[int(0.90*(len(nets)-1))]

    return {
        "risk_score": s,
        "risk": risk,
        "hold_used": hold,
        "mean": mean,
        "median": median,
```

```
        "p10": p10,
        "p90": p90
    }

    ##The Plot
import matplotlib.ticker as mtick

MALE_COLOR  = "#2563EB"
FEMALE_COLOR = "#DB2777"
BAR_WIDTH   = 0.36

x = np.arange(len(AGES))

male_means  = [results[("Male", a)]["mean"] for a in AGES]
female_means = [results[("Female", a)]["mean"] for a in AGES]
male_p10    = [results[("Male", a)]["p10"] for a in AGES]
male_p90    = [results[("Male", a)]["p90"] for a in AGES]
female_p10  = [results[("Female", a)]["p10"] for a in AGES]
female_p90  = [results[("Female", a)]["p90"] for a in AGES]


plt.rcParams.update({
    "font.size": 11,
    "axes.titlesize": 14,
    "axes.labelsize": 12,
    "axes.titleweight": "bold",
})

fig, ax = plt.subplots(figsize=(10.5, 6.5), dpi=140)

fig.patch.set_facecolor("white")
ax.set_facecolor("#F8FAFC")

ax.grid(axis="y", alpha=0.25, linewidth=1, zorder=0)
ax.set_axisbelow(True)

# Bars
bars_m = ax.bar(x - BAR_WIDTH/2, male_means, BAR_WIDTH,
               color=MALE_COLOR, alpha=0.88, label="Male", zorder=3)
bars_f = ax.bar(x + BAR_WIDTH/2, female_means, BAR_WIDTH,
               color=FEMALE_COLOR, alpha=0.88, label="Female", zorder=3)

# Whiskers: p10p90
whisk_c = (0.15, 0.23, 0.35, 0.65) # RGBA slate-ish
capw = 0.06

for i in range(len(AGES)):
    # Male
    ax.plot([x[i] - BAR_WIDTH/2] * 2, [male_p10[i], male_p90[i]],
            color=whisk_c, lw=2.0, solid_capstyle="round", zorder=4)
    ax.plot([x[i] - BAR_WIDTH/2 - capw, x[i] - BAR_WIDTH/2 + capw],
            [male_p10[i], male_p10[i]], color=whisk_c, lw=1.6, zorder=4)
    ax.plot([x[i] - BAR_WIDTH/2 - capw, x[i] - BAR_WIDTH/2 + capw],
            [male_p90[i], male_p90[i]], color=whisk_c, lw=1.6, zorder=4)

    # Female
    ax.plot([x[i] + BAR_WIDTH/2] * 2, [female_p10[i], female_p90[i]],
            color=whisk_c, lw=2.0, solid_capstyle="round", zorder=4)
    ax.plot([x[i] + BAR_WIDTH/2 - capw, x[i] + BAR_WIDTH/2 + capw],
```

```
            [female_p10[i], female_p10[i]], color=whisk_c, lw=1.6, zorder=4)
    ax.plot([x[i] + BAR_WIDTH/2 - capw, x[i] + BAR_WIDTH/2 + capw],
            [female_p90[i], female_p90[i]], color=whisk_c, lw=1.6, zorder=4)

# Break-even line (more visible, still subtle)
ax.axhline(0, color="#64748B", lw=1.2, ls="--", zorder=2, alpha=0.9)

# Format y-axis as dollars
ax.yaxis.set_major_formatter(mtick.FuncFormatter(lambda v, _: f"${v:,.0f}"))

# X axis labels
ax.set_xticks(x)
ax.set_xticklabels(AGES)

ax.set_xlabel("Age Group")
ax.set_ylabel("Expected Annual Net ($)")

ax.set_title(
    "Avg. Annual Net Outcome (Active Gamblers)\n"
    ,
    color="#0F172A", pad=12
)
##"Bars show mean net; whiskers show 10th90th percentile range"

ax.spines["top"].set_visible(False)
ax.spines["right"].set_visible(False)
ax.spines["left"].set_color("#CBD5E1")
ax.spines["bottom"].set_color("#CBD5E1")

leg = ax.legend(loc="lower right", frameon=True)
leg.get_frame().set_facecolor("white")
leg.get_frame().set_edgecolor("#CBD5E1")
leg.get_frame().set_alpha(0.95)

def add_labels(bar_container, color_text="#0F172A"):
    for b in bar_container:
        h = b.get_height()
        if np.isnan(h):
            continue
        x0 = b.get_x() + b.get_width() / 2
        # Place slightly above for positive, slightly below for negative
        y0 = h + 25 if h >= 0 else h - 35
        va = "bottom" if h >= 0 else "top"
        ax.text(x0, y0, f"${h:,.0f}", ha="center", va=va,
                fontsize=9.5, fontweight="bold", color=color_text, zorder=5)

add_labels(bars_m)
add_labels(bars_f)

caption = (
    f"Monte Carlo simulation (n={N_SIM} simulated people per agegender group; active gamblers
        only).\n"
    "Hold by risk tier: Low=8%, Medium=10%, High=15%. Education fixed at BA+ in this run."
)
fig.text(0.5, 0.01, caption, ha="center", fontsize=9, color="#475569")

plt.tight_layout(rect=[0, 0.05, 1, 1])
plt.show()
```

```python

# Testing Data
if __name__ == "__main__":
    random.seed(7)

    example_profile = {
        "gender": "male",
        "age": "18-34",
        "edu": "ba_plus",
        "has_account": True,
        "placed_bet": True,
        "bet_freq": "3_4_week",
        "deposit_freq": "weekly_plus",
        "withdraw": "leave_in_account",
        "chases": True,
        "annual_handle_est": 4500.0 # pretend Step 3 estimated this
    }

    out = simulate_annual_net(example_profile, n_trials=8000)

    print("=== Annual Net Prediction ===")
    print(f"Risk tolerance: {out['risk']} (score {out['risk_score']})")
    print(f"Effective hold used: {out['hold_used']:.3f}")
    print(f"Annual handle estimate: ${example_profile['annual_handle_est']:,.0f}")
    print(f"Expected annual net (mean): ${out['mean']:,.0f}")
    print(f"Median annual net: ${out['median']:,.0f}")
    print(f"10th90th  percentile net: ${out['p10']:,.0f} to ${out['p90']:,.0f}")
```

7.3 Don't Break the Bank

```python

#Set up file mount from google drive

from google.colab import drive
drive.mount('/content/drive')
CSV_PATH = '/content/drive/MyDrive/m3challenge/sb_gdp_impact_US_UK.csv'

#imports
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import matplotlib.ticker as mtick
import seaborn as sns
sns.set(style='whitegrid')


# Ensure numeric columns & check to make sure google drive properly mounted

numcols = ['VA_gambling','GGR_gambling_total','GGR_sports','VA_sports','GDP','Population']
for c in numcols:
    if c in df.columns:
        df[c] = pd.to_numeric(df[c], errors='coerce')

df


```

```python

# VA_sports_model formular implementation
df['VA_sports_model'] = df['VA_gambling'] * (df['GGR_sports'] / df['GGR_gambling_total'])
df['GDPShare_pct_model'] = 100.0 * df['VA_sports_model'] / df['GDP']
df['PerCapitaVA_model'] = df['VA_sports_model'] / df['Population']

#round each value to 6 decimal places
df.round(6)


#Economic Impact Model implementation

def simulate_fiscal_effect(df, tau=0.20, mg=1.2, lam=0.6, alpha=None):
    d = df.copy()
    if alpha is not None:
        d['DirectVA'] = alpha * d['GGR_sports']
    else:
        d['DirectVA'] = d['VA_sports_model']
    d['TaxRevenue'] = tau * d['GGR_sports']
    d['FiscalDeltaGDP'] = (mg - 1.0) * d['TaxRevenue']
    d['NetDeltaGDP_pre_sub'] = d['DirectVA'] + d['FiscalDeltaGDP']
    d['NetDeltaGDP'] = (1.0 - lam) * d['NetDeltaGDP_pre_sub']
    d['NetGDPShare_pct'] = 100.0 * d['NetDeltaGDP'] / d['GDP']
    d['NetPerCapitaUSD'] = d['NetDeltaGDP'] / d['Population']
    return d

# Baseline parameters for formula
tau = 0.20  # tax rate (20%)
mg = 1.2    # fiscal multiplier
lam = 0.6   # substitution (60% diverted from other spending)

df_sim = simulate_fiscal_effect(df, tau=tau, mg=mg, lam=lam)
df_sim[['country','VA_sports_model','TaxRevenue','FiscalDeltaGDP','NetDeltaGDP','NetGDPShare_pct','NetPerCapit


# Plot baseline Net GDP share with per-capita line
import matplotlib.pyplot as plt

df_plot = df_sim.sort_values('NetGDPShare_pct', ascending=False).reset_index(drop=True)
fig, ax = plt.subplots(figsize=(9,5))
bars = ax.bar(df_plot['country'], df_plot['NetGDPShare_pct'], color=['#2b7bba','#c94b4b'],
    edgecolor='black')
ax.set_ylabel('Estimated net sports-betting contribution (% of GDP)')
ax.yaxis.set_major_formatter(mtick.PercentFormatter())
ax.set_title(f'Net GDP impact (tau={tau:.0%}, mg={mg}, lambda={lam})')

# annotate bars
for bar, v in zip(bars, df_plot['NetGDPShare_pct']):
    ax.annotate(f'{v:.4f}%', xy=(bar.get_x()+bar.get_width()/2, v), xytext=(0,6),
        textcoords='offset points', ha='center')

ax2 = ax.twinx()
ax2.plot(df_plot['country'], df_plot['NetPerCapitaUSD'], marker='o', linestyle='-',
    color='black')
ax2.set_ylabel('Net per-capita sports-betting VA (USD)')
for x, y in zip(df_plot['country'], df_plot['NetPerCapitaUSD']):
    ax2.annotate(f'${y:,.0f}', xy=(x, y), xytext=(0, -12), textcoords='offset points',
        ha='center', va='top', fontsize=9)

```

```
plt.tight_layout()
plt.show()

import pandas as pd
import numpy as np
import matplotlib.pyplot as plt


# tax rates
US_tax_rate = 0.20
UK_tax_rate = 0.15

# compute tax revenue
def calc_tax(row):
    if row["country"] == "US":
        return row["GGR_sports"] * US_tax_rate
    else:
        return row["GGR_sports"] * UK_tax_rate

df["TaxRevenue"] = df.apply(calc_tax, axis=1)

# plotting + labels
labels = df["country"]
gross = df["GGR_sports"]
tax = df["TaxRevenue"]

x = np.arange(len(labels))
width = 0.35

plt.figure(figsize=(7,5))
plt.bar(x - width/2, gross, width, label="Sportsbook Gross Revenue")
plt.bar(x + width/2, tax, width, label="Government Tax Revenue")

plt.xticks(x, labels)
plt.ylabel("USD")
plt.title("Government Capture of Sports Betting Revenue")
plt.legend()
plt.grid(axis='y', linestyle='--', alpha=0.4)

plt.tight_layout()
plt.show()


#Sensitivity heatmap for a selected country
#US shown
import seaborn as sns

country = 'US'
mg_fixed = 1.2
sub = sweep[(sweep['country']==country) & (sweep['mg']==mg_fixed)]
pivot = sub.pivot(index='tau', columns='lam', values='NetGDPShare_pct')

plt.figure(figsize=(7,4))
sns.heatmap(pivot*100, annot=True, fmt='.4f', cmap='viridis') # convert to percent

#set up graph labels
plt.title(f'Sensitivity of Net GDP share (%)  {country} (mg={mg_fixed})')
plt.ylabel('Tax rate (tau)')
plt.xlabel('Substitution (lambda)')
```

```
plt.show()
```